

Completing WebWork Assignments

Over half of your homework grade will come from homework assignments you will complete in WebWork. These problems will mostly come from the even numbered problems in the textbook, sometimes with randomly generated numbers so that different people will have slightly different problems.

To work on an assignment, use the link sent to your ttu email. Your username is your eraider name, but your initial password was sent to your ttu.edu email address. (For most people this password will be your R number.) Be sure to try logging in as soon as possible and let me know if you have any trouble. You can email me at brock.williams@ttu.edu. Once you've logged on, you should change your password to something a bit more secret, but be sure that it's something you will remember, like your eraider password.

Next, click on "Begin Problem Sets." You will notice that there are two available problem sets. The first, Set00, is just for practice and doesn't count in your grade. It explains how to type your answers and use the system.

I highly recommend using the "Download a hardcopy" link to download and print the problems before trying them online. Then you can work on them even if you are away from the computer. Most people will want to choose PDF instead of TeX.

You can answer the questions as many times as you like before the deadline (except for multiple choice and true/false questions, of course). So don't worry if you get an answer wrong, it won't count against you so long as you get it right before the deadline. Also, once you get a question right, it is marked as correct and you will get credit for it, even if you accidentally change your answer to something that's wrong. That means that you can re-work the homework before the exams without worrying about messing up your grade.

With only a very few exceptions, you can type your answer exactly the same way you would write it or type it into the calculator. For example, it knows that 8, $2+6$, and $16/2$ are all the same answer. To see if it understands what you've typed, try hitting the "Preview Answers" button before hitting "Submit Answers." It will show you what you typed and what it thinks you meant. If there are mistakes in your answer, for example, if the answer should be a number, like 8, and you type $\sin(x)$, then it will tell you that you're giving the wrong kind of answer. There are a few things to be aware of however:

1. If you don't use parentheses correctly, WebWork might misunderstand what you type. For example, WebWork thinks $\sin x$ is the same as $\sin(x)$ because it assumes the first thing after the sin is what you want to take the sine of. But if you type $\sin 2x$, then it thinks you mean $\sin(2) * x$. When in doubt, be sure to "Preview Answers" so that you know what WebWork thinks you mean.
2. The only calculation that WebWork does incorrectly is finding an odd root of a negative number. For example, the cube root of -8 is, of course, -2. But if you enter $(-8)^{(1/3)}$, the preview will announce "Can't raise a negative number to a power." The problem, as we'll see later, is that the

right way to interpret $(-8)^{1/3}$ is $e^{1/3 \ln(-8)}$. But the underlying Perl function for $\ln$ can't find the natural log of a negative number. The solution for us is to just enter -2 directly, or even $(-8)^{1/3}$.

3. Be sure to use 4 or 5 digits for long decimal answers so that WebWork believes you got the right answer. For example, if the answer is $1/3$, you can type $1/3$ or 0.333333 , but not just 0.3 .
4. You can use π or e to mean the constants π or e .
5. You can use either $\wedge$ or $**$ to represent exponentiation. For example, you can enter x squared as x^2 or as $x**2$. But WebWork will print the answers using the double star notation.
6. WebWork understands expressions like $\sin^2 x$ or $\sin^{(-1)} x$, but when it prints answers, it will use function names like $\text{asin}(x)$, $\text{acos}(x)$ for the inverse trig functions.
7. Be sure to read the question carefully. For example, some questions will ask you to type “inf” for infinity, others will ask you to spell out infinity completely.
8. If WebWork still seems to misunderstand you, there is a link to a quick reminder of the rules on the page for each problem set. Or you can always use the “Email instructor” button at the bottom of almost every page to send me an email explaining what's wrong. Just remember to start working on the assignment well in advance of the deadline. Since assignments will be due at 11:59 pm, if you email me too late, I can't promise that I'll be able to respond in time.